

**LAPORAN PRAKTIKUM  
GRAFIKA KOMPUTER  
POSTTEST 4**



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**PROGRAM STUDI INFORMATIKA  
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## PEMBAHASAN

Aplikasi **Model Kit Hobby** adalah aplikasi e-commerce sederhana berbasis Flutter yang menjual berbagai model kit (Gundam, tank, pesawat, dan kapal). Aplikasi ini menggunakan `Image.asset`, `Stack`, `Positioned`, `BoxShadow`, `Navigator.push`, `Navigator.pop`, `inputFormatters`, `NavigationBar`, serta `StatefulWidget` dengan `setState()`.

**Beranda** — menampilkan judul toko, kolom pencarian real-time, tombol filter, dan grid produk 3 kolom berisi gambar, badge ukuran, tombol tambah, nama, dan harga.

**Keranjang** — menampilkan produk yang sudah ditambahkan, input qty (hanya angka), total harga otomatis, dan tombol Checkout.

**Profil** — menampilkan avatar, nama, email, dan daftar menu.

**Detail Produk** — terbuka saat produk di-tap dari Beranda, menampilkan gambar besar, deskripsi, harga, dan tombol aksi.

**Checkout** — halaman konfirmasi yang terbuka dari tombol Checkout di Keranjang, menampilkan ikon centang, total, dan tombol Konfirmasi.

Navigasi antar halaman utama menggunakan **NavigationBar** (3 tab), sedangkan halaman Detail dan Checkout dibuka dengan **Navigator.push** dan kembali dengan **Navigator.pop**.

### Widget

1. **MaterialApp** — `main.dart` — Root widget aplikasi.
2. **ThemeData** — `main.dart` — Konfigurasi tema aplikasi.
3. **StatefulWidget** — `main.dart`, `home_page.dart`, `cart_page.dart` — Widget yang memiliki state dinamis.
4. **StatelessWidget** — `product_card.dart`, `detail_page.dart`, `checkout_page.dart`, `profile_page.dart` — Widget tanpa state.
5. **Scaffold** — `main.dart`, `detail_page.dart`, `checkout_page.dart` — Kerangka halaman.
6. **NavigationBar** — `main.dart` — Navigasi bawah 3 tab.
7. **NavigationDestination** — `main.dart` — Item tab (icon + label).
8. **Badge** — `main.dart` — Jumlah item keranjang di tab.
9. **SafeArea** — `home_page.dart`, `cart_page.dart`, `profile_page.dart` — Menghindari area sistem.
10. **SingleChildScrollView** — `home_page.dart`, `detail_page.dart` — Konten bisa di-scroll.
11. **Padding** — Semua halaman — Memberi jarak widget.

12. **Column** — Semua halaman — Menyusun widget vertikal.
13. **Row** — home\_page.dart, cart\_page.dart, detail\_page.dart, checkout\_page.dart — Menyusun widget horizontal.
14. **Center** — home\_page.dart, product\_card.dart, detail\_page.dart, checkout\_page.dart — Menempatkan widget di tengah.
15. **Text** — Semua halaman — Menampilkan teks.
16. **SizedBox** — Semua halaman — Jarak & ukuran.
17. **Expanded** — home\_page.dart, cart\_page.dart, product\_card.dart, detail\_page.dart — Mengisi ruang sisa.
18. **TextField** — home\_page.dart, cart\_page.dart — Input pencarian & qty.
19. **InputDecoration** — home\_page.dart, cart\_page.dart — Dekorasi TextField.
20. **OutlineInputBorder** — home\_page.dart, cart\_page.dart — Border rounded.
21. **FilteringTextInputFormatter** — cart\_page.dart — Batasi input angka.
22. **LengthLimitingTextInputFormatter** — cart\_page.dart — Batasi panjang input.
23. **TextEditingController** — cart\_page.dart — Kontrol nilai TextField.
24. **Container** — Semua halaman — Kotak pembungkus.
25. **BoxDecoration** — Semua halaman — Dekorasi background & border.
26. **BoxShadow** — product\_card.dart, home\_page.dart, cart\_page.dart, detail\_page.dart, profile\_page.dart — Bayangan widget.
27. **Border** — product\_card.dart — Border card.
28. **BorderRadius** — home\_page.dart, cart\_page.dart, profile\_page.dart — Sudut melengkung.
29. **GridView.builder** — home\_page.dart — Grid produk 3 kolom.
30. **SliverGridDelegateWithFixedCrossAxisCount** — home\_page.dart — Atur kolom grid.
31. **ListView.builder** — cart\_page.dart — Daftar produk keranjang.
32. **ListView** — profile\_page.dart — Daftar menu profil.
33. **Stack** — product\_card.dart, detail\_page.dart, profile\_page.dart — Menumpuk widget.
34. **Positioned** — product\_card.dart, detail\_page.dart, profile\_page.dart — Atur posisi di Stack.

35. **Positioned.fill** — product\_card.dart, detail\_page.dart — Isi seluruh area Stack.
36. **Image.asset** — product\_card.dart, detail\_page.dart, cart\_page.dart — Gambar dari assets.
37. **Icon** — Semua halaman — Menampilkan ikon.
38. **IconButton** — detail\_page.dart, checkout\_page.dart — Tombol ikon kembali.
39. **GestureDetector** — product\_card.dart — Deteksi tap card & tombol tambah.
40. **CircleAvatar** — profile\_page.dart — Avatar bulat pengguna.
41. **ListTile** — profile\_page.dart — Baris menu profil.
42. **AppBar** — detail\_page.dart, checkout\_page.dart — Header halaman.
43. **OutlinedButton.icon** — detail\_page.dart — Tombol outline.
44. **ElevatedButton.icon** — detail\_page.dart, cart\_page.dart — Tombol solid dengan ikon.
45. **ElevatedButton** — checkout\_page.dart — Tombol konfirmasi.
46. **SnackBar** — cart\_page.dart, checkout\_page.dart — Notifikasi bawah.
47. **ScaffoldMessenger** — cart\_page.dart, checkout\_page.dart — Menampilkan SnackBar.
48. **Navigator.push** — product\_card.dart, cart\_page.dart — Pindah halaman.
49. **Navigator.pop** — detail\_page.dart, checkout\_page.dart — Kembali halaman.
50. **MaterialPageRoute** — product\_card.dart, cart\_page.dart — Route navigasi.
51. **setState()** — main.dart, home\_page.dart, cart\_page.dart — Update UI saat state berubah.
52. **initState()** — cart\_page.dart — Lifecycle pertama, inisialisasi controller.
53. **didUpdateWidget()** — cart\_page.dart — Lifecycle saat parent rebuild.
54. **dispose()** — cart\_page.dart — Lifecycle terakhir, bersihkan controller.
55. **onChanged** — home\_page.dart, cart\_page.dart — Callback saat user mengetik.
56. **Callback Function** — home\_page.dart (onTambah), cart\_page.dart (onUbahQty, onQuantityChanged) — Kirim perubahan ke parent.
57. **State Turunan (getter)** — main.dart (\_totalItems), home\_page.dart (visibleProducts), cart\_page.dart (grandTotal, items) — Nilai dihitung dari state.
58. **Conditional Rendering** — home\_page.dart, cart\_page.dart — Widget berbeda sesuai kondisi.

59. **onPressed conditional** — cart\_page.dart — Tombol Checkout aktif jika total > 0.

Link video

<https://drive.google.com/drive/folders/1ScZlhDbpl2fNGIZtuvGrUILOoIR0NSCD?hl=ID>